Paper review · Visionary

Teacher recoverability in DAgger for VLAs

DAgger: Ross, Gordon and Bagnell, AISTATS 2011 (arXiv:1011.0686).

Pavel Bushuyeu · October 2026

Which states should a VLA teacher label?

DAgger assumes a good expert

Theorem 3.2: J(DAgger policy) ≤ J(expert) + u·T·ε + O(1). u bounds the extra cost of one learner action followed by the expert.

The bound is linear in T when u stays bounded, for example when the expert recovers in a few steps. A VLA teacher can fail from the student’s states.

Two recent selectors

Kintsugi-VLA measures recoverability from saved states, with a privileged expert. PIVOT-Q labels only states where teacher and student disagree.

I found no study that compares the two for a VLA teacher on student states.

Hypotheses

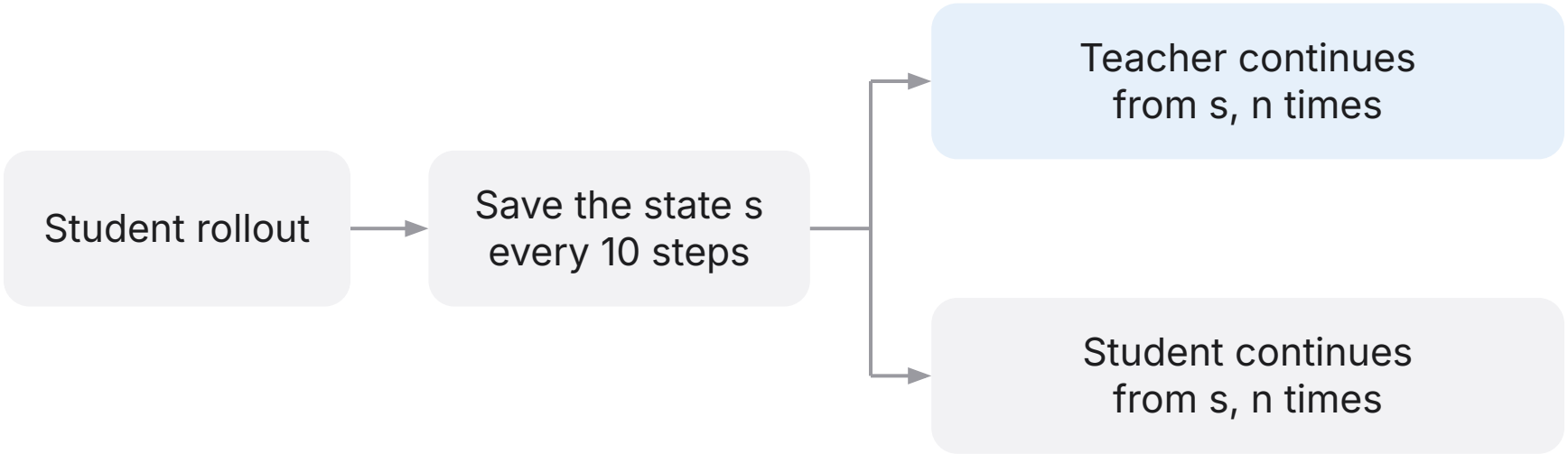
H1 From the student's states, the teacher finishes the task less often than from matched start states.

H2 At the same label count and total cost, recoverable states give better labels than high-disagreement states or all states.

H3 The gain from on-policy labels over offline distillation grows with the student's imitation error.

If H1 holds, the labels of a VLA teacher need a check before use. If H2 holds, the check is worth its cost.

Measuring teacher recoverability



r(s) is the share of teacher continuations that finish the task in the remaining steps, with a Wilson interval. The student continuations are the control.

Design

Teacher	π0.5 LIBERO checkpoint (openpi), frozen.
Student	SmolVLA (450M), round 0 on 10% and on 50% of the demos.
Selectors	All states (DAgger); disagreement; recoverability; random.
Recovery	n = 10 continuations per state; success within the step budget.
Budgets	Same labeled chunks; separately, same total cost.
Tasks	LIBERO, 4 suites; LIBERO-Plus camera and layout shifts.
Metrics	Success vs labels and vs cost; r(s) by task and time.

Compute and risks

Compute (estimate)

Per round: 300 tested states, 10 teacher and 10 student continuations each. Three rounds: 18,000 continuations.

With training and evaluation, about 200 GPU-hours on one RTX 5090. The pilot measures the real speed first.

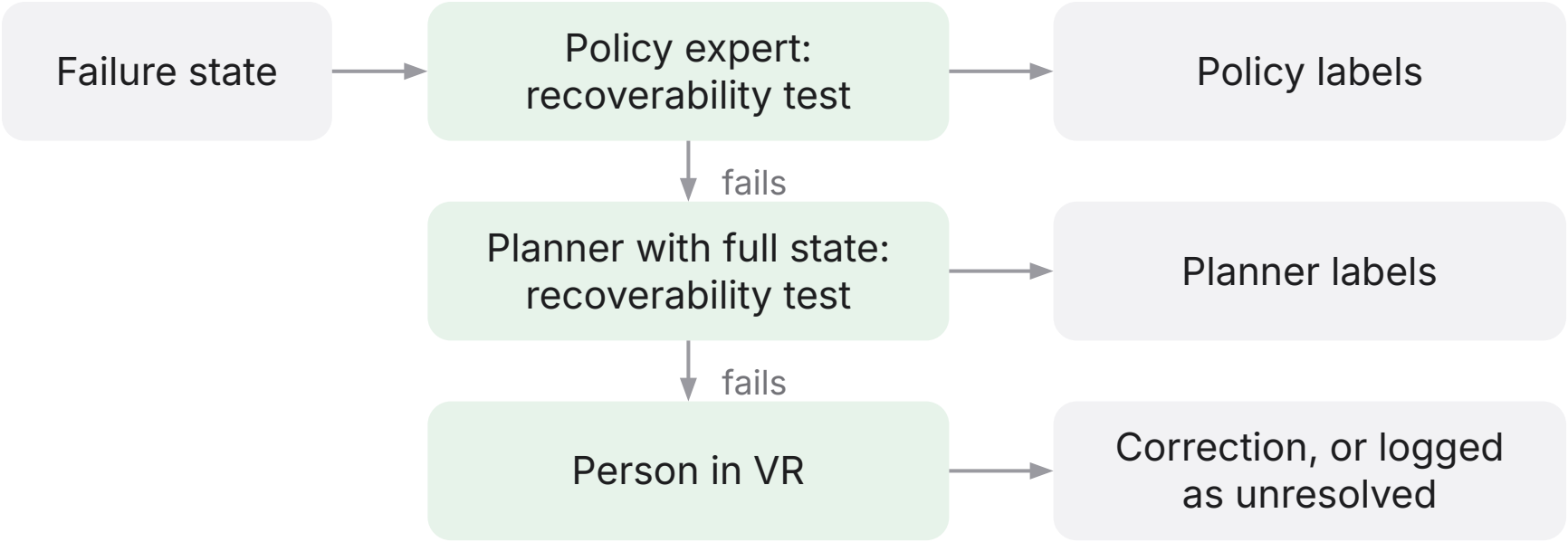
Risks

The teacher is strong (92.4% on LIBERO-Long), so few states may be unrecoverable.

The restore must include controller state and history. The pilot checks replay.

PIVOT-Q already selects by disagreement. The new part is recoverability.

Application: a failure router



A failure goes to the cheapest tested expert that passes the threshold.

Outputs and prior work

For simulated household tasks

- Recovery data for each failure type.
- Human time only after the automated experts fail.
- A table per task: which expert handles which failures.

Prior work

Kintsugi-VLA picks recovery starts with one privileged expert. HG-DAgger and RaC use a person for each correction.

The 2025 BEHAVIOR winner had no recovery demonstrations. It used hand-written correction rules.

Pilot first, then the router

Pilot (go/no-go)

LIBERO-Long, the round-0 student, 300 saved states by task, time and outcome.

Go on if the restore replays exactly and r(s) varies between states. A high r(s) everywhere is a negative result for H1.

Router test

2–3 household tasks in simulation, with a policy expert, a planner with full state and a person in VR.

Policy-only, human-only and the router at the same cost: student success and human minutes.

What this builds on

Work	What it does	What this proposal adds
DAgger, 2011	Expert labels all states; bound has u.	A recovery test for the expert.
Kintsugi-VLA, 2026	Recoverability with a privileged expert.	A VLA teacher; DAgger labels.
PIVOT-Q, 2026	Labels only high-disagreement states.	Recoverability as selector.
VLA-OPD, 2026	A VLA teacher labels student states.	A check of those labels.
HG-DAgger, RaC	A person takes over and corrects.	People only after models fail.
Reward-DAgger, 2026	A reward model gates π0.5 takeovers.	Which states π0.5 can fix.

Sources

1. Ross, Gordon and Bagnell. DAgger, AISTATS 2011: Theorems 2.2 and 3.2.
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4. Kelly et al. HG-DAgger, 2019. Hu et al. RaC, 2025.
5. Foster, Block and Misra. Is Behavior Cloning All You Need?, 2024.
6. Physical Intelligence. π0.5, 2025, and the openpi LIBERO checkpoint. Shukor et al. SmolVLA, 2025.
7. Liu et al. LIBERO, 2023. LIBERO-Plus, 2025.
8. Larchenko et al. 1st place, 2025 BEHAVIOR Challenge, 2025.